A Short Academic Project Proposal On

IoT-Based Smart Agriculture Monitoring and Decision Support System

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Supervisor Endorsement

We certify that the students, as final-year candidates of the Bachelor of Computer Engineering program, possess the required academic preparedness, technical competence, and research aptitude to successfully undertake this work as part of their major project. The proposed research reflects a comprehensive application of core Computer Engineering principles, including programming, data processing, embedded systems, networking, and intelligent computing, as covered throughout their academic curriculum.

The study is appropriate to their level of study and demonstrates the effective integration of theoretical knowledge with practical implementation to address real-world challenges. Furthermore, the research exhibits relevance to contemporary technological and environmental concerns by promoting efficient resource utilization and sustainable practices. The project aligns well with our areas of academic expertise, enabling us to provide effective supervision, technical guidance, and scholarly oversight to ensure the academic quality, feasibility, and relevance of the work.

Supervisor(s):

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Er. Kamal Lekhak

Date: February 11, 2026

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1. Introduction

1.1 Background

Agriculture contributes to about one-third of gross domestic product (GDP) in Nepal reaching nearly Rs 57.48 billion in the current fiscal year 2082/83 BS and provides employment to 74% of the economically active population.

Despite this, the agricultural sector still faces challenges from low productivity, traditional methods, inadequate access to advanced technologies, and increasing climate change sensitivity. Climate change has brought unpredictable hailstorms, floods, and unseasonal rains that destroy crops and reduce yields. Farmers often rely on intuition and other irrelevant factors, such as the desire for quick profits or ignorance of market demands, when deciding which crops to cultivate. This can lead to poor choices and financial strain on farmers, leading to discouragement from the profession.

Agriculture remains the backbone of many economies, yet it faces unprecedented challenges: unpredictable weather, water shortage, soil degradation, pest attacks, and lack of timely expert advice. According to the United Nations, food production must increase by 60–70% by 2050 to feed 10 billion people. Traditional methods rely heavily on experience and guesswork, leading to over-irrigation (wasting up to 50% water), improper fertilizer use, and delayed response to crop stress.

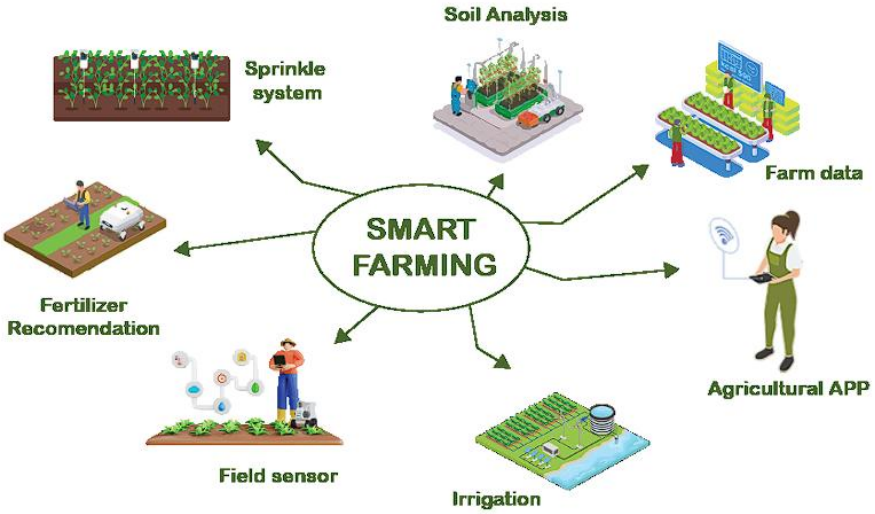

Figure 1: Smart Farming (soil analysis, irrigation system, fertilizer recommendation) [8].

The convergence of modern technology and renewable resources offers a transformative solution. Low- cost sensors powered with solar energy can continuously monitor soil and weather parameters and provide a sustainable system and machine learning models can predict optimal actions with high accuracy. This project implements exactly such an integrated system.

1.2 Problem Statement

Problems in the current agriculture system are:

- Dependency on weather and unverified information.
- Intuition-based decisions.
- Lack of real time field data.
- Illiteracy about soil characteristics.
- Reduced crop productivity.

These issues make traditional farming less productive, less profitable, and less sustainable in Nepal. So, our proposed system will mitigate these problems and help farmers to make data driven decisions rather than intuition based imprecise decisions.

1.3 Objectives

The main objectives of this project are:

- To implement ML models to analyze data, and provide recommendations for crops, fertilizers and irrigation.
- To build an IoT based system with a user-friendly decision support dashboard
- To make farming smart and sustainable.
- To make use of renewable energy (solar based power supply) for efficient system operation.
- To ensure optimization of water and fertilizing resources.

2. Research Methodology

2.1 Literature Review

The integration of Internet of Things (IoT), cloud computing, and artificial intelligence (AI) to transform traditional agriculture into a fully automated, data-driven system has gained significant momentum in recent years.

Alreshidi highlighted the fragmentation issues in current agricultural processes and proposed a comprehensive Smart Sustainable Agriculture (SSA) platform underpinned by IoT and AI that addresses sensor data collection, cloud storage, big data analytics, interoperability, and real-time decision support – a vision that directly aligns with the architecture of the proposed system [1].

Quy et al. presented a detailed survey of IoT-enabled smart agriculture ecosystems, emphasizing layered architecture comprising perception (sensors/WSN), network, cloud processing, big data analytics, and application layers. They stressed the necessity of real-time monitoring of soil moisture, temperature, humidity, pH, nutrient levels, and weather data, and pest detection, followed by AI-driven predictive analytics and farmer-facing interfaces – essentially the exact blueprint this project follows [2].

The fusion of Wireless Sensor Networks, IoT, Cloud Computing, and AI/ML for smart farming, proposing a modular architecture that includes (i) WSN for data acquisition, (ii) edge/cloud processing, (iii) AI models for crop recommendation, irrigation scheduling, yield prediction, and crop rotation planning, and (iv) user-friendly web/mobile interfaces for real-time visualization and control. The authors explicitly state that such integrated systems lead to higher crop quality, optimal resource use, and weather-responsive strategies, validating the core objectives of the present work [3].

Ibrahim et al. demonstrated the practical feasibility of combining IoT with deep learning (ResNet50 achieving 99.8% accuracy) for automated plant disease detection and treatment using low-cost hardware mounted on existing irrigation pivots, proving that high-accuracy AI models can run locally or via cloud with minimal latency and provide immediate

actionable outputs (e.g., targeted spraying), which strengthens the case for including pest/disease prediction modules in the proposed platform [4].

Hossain et al. investigated an IoT-based smart irrigation system powered by renewable energy, specifically focusing on solar energy integration to improve water and energy efficiency in agricultural contexts. Their work highlights how sensor networks can actively monitor soil moisture and environmental parameters and, when combined with intelligent control logic, automate the irrigation process to reduce water wastage. The study demonstrated that solar-powered sensor systems not only lower operational costs but also enhance system sustainability for remote and smallholder farms where grid electricity is unreliable. The outcomes underscore the potential of renewable energy-driven IoT solutions to support climate-adaptive irrigation strategies, an insight that strengthens the design rationale of the proposed project's renewable-powered decision support system [5].

The study by Kim et al. focuses on the application of IoT sensors paired with machine learning models to monitor soil and environmental conditions that support improved crop decision-making. The research emphasizes the importance of real-time data streams, including soil moisture, temperature, and nutrient levels, in enabling proactive irrigation scheduling and crop recommendations. Additionally, this paper points to scalable system design and adaptability across different agricultural settings, which reinforces the feasibility of deploying data-driven agricultural systems in smallholder and resource-constrained environments. The authors' validation of predictive analytics strategies directly informs the machine learning components of the current proposal [6].

Reddy and Ghosh explored the design and performance of IoT networks specifically applied to smart farming environments, with detailed assessment of communication protocols, sensor integration strategies, and data reliability metrics. Their findings emphasize that proper network architecture, including low-power wide-area technologies, is crucial for consistent data acquisition in fields with irregular power and connectivity conditions. By evaluating different architectural strategies, this study highlights key design considerations that influence the responsiveness and scalability of IoT-based agricultural systems insights that are directly applicable to the hardware and communication layer design of the proposed project [7].

Zhang et al. presented an advanced IoT agriculture platform that integrates edge computing with cloud-based analytics for real-time decision support. The study demonstrated optimized data processing through distributed computing, enabling faster response times for irrigation and crop recommendations under varying environmental conditions. Their architecture addressed challenges related to data latency, computational overhead, and energy consumption by balancing workload between edge devices and cloud servers. This hybrid computing approach is particularly relevant to the current proposal, which aims to use both local processing and cloud services to provide actionable recommendations without overwhelming limited hardware resources [8].

Liu and Singh examined the socio-technical challenges of adopting IoT systems in rural agriculture, including affordability, farmer literacy, and technology trust issues. Their research highlights that while IoT infrastructures can significantly improve farming outcomes, successful adoption depends on user-centered design, local adaptability, and intuitive interfaces that bridge the gap between complex analytics and practical field decisions. The authors also note the importance of affordable hardware and renewable power options to increase uptake among smallholder communities. These insights directly support the proposed project's focus on low-cost, solar-powered systems and emphasize the need for user-centric dashboard design [9].

Ahmed and Khan investigate the deployment of machine learning models for precision agriculture, focusing on predictive crop yield estimation and soil health analysis using sensor networks. Their results demonstrated that regression and ensemble learning models significantly outperform traditional rule-based systems in predicting optimal crop cycles and fertilizer recommendations. Additionally, the study discusses the value of continuous data streams in training adaptive models that improve over time as more environmental data is collected. These findings provide strong empirical support for implementing machine learning techniques within the current project's decision support framework [10].

These studies collectively confirm that the convergence of IoT sensors, cloud-based storage and analytics, AI/ML for predictive insights (crop suitability, irrigation/fertilization scheduling, yield forecasting), and intuitive web/mobile dashboards represents the current

state-of-the-art and the most promising pathway to sustainable, profitable, and climate-resilient agriculture – precisely the foundation upon which this project is built.

2.2 Requirement Specification

2.2.1 Functional Requirements

- Real-time collection of soil and environmental parameters, including pH, soil moisture, temperature, and humidity, using IoT sensors.
- Transmission of sensor data to a FastAPI backend via WiFi (ESP32) or LoRa for long-range communication.
- Storage of raw sensor readings and processed predictions in a database.
- Execution of ML models to provide crop recommendations and fertilizer suggestions based on soil and weather data.
- Alerts and notifications for soil anomalies, crop suitability, and irrigation/fertilizer needs.
- Visualization of live sensor readings, predictions, and historical data through a frontend dashboard accessible to farmers.

2.2.2 Non-Functional Requirements

- Reliability: Continuous and accurate sensor monitoring.
- Scalability: Support multiple sensors and farms without performance loss.
- Real-time Performance: Quick sensor processing and ML predictions.
- Usability: Clean, user-friendly dashboard for farmers.
- Maintainability: Easy updates for ML models, sensors, and features.
- Availability: High uptime with minimal downtime.

2.2.3 System Requirements

Software & Technologies

- Firmware : Arduino IDE / MicroPython
- Cloud : ThingSpeak (free tier) or AWS IoT Core + DynamoDB
- Backend : Python FastAPI or Node.js
- Database : MongoDB / Firebase Realtime DB

- AI/ML : Python (scikit-learn, TensorFlow/Keras)
- Frontend : React.js + Shadcn
- Communication: MQTT protocol or HTTPS protocol

The approximate hardware requirement is described in the table below:

Table 1: Hardware requirements description

Component	Purpose	Justification	Possible Replacements / Complements
ESP32 Microcontroller	Collects sensor data and transmits it to backend	Low cost, built-in WiFi/Bluetooth, easy to program	Arduino Uno/Nano (for simpler setups), Raspberry Pi Pico W (more computational power)
pH Sensor	Measures soil acidity for crop recommendation	pH affects which crops can grow	Alternative: Atlas Scientific pH probe (more precise)
Soil Moisture Sensor	Monitors water content for irrigation recommendations	Moisture impacts crop growth	Capacitive sensors preferred over resistive sensors for accuracy
BME 280 (Temperature & Humidity)	Measures environmental conditions	Temperature & humidity affect crop suitability	DHT11 or DHT22 (cheaper, less precise)
NPK Sensor	Measures soil nutrient levels (Nitrogen, Phosphorus, Potassium)	Necessary for fertilizer recommendations	DIY ion-selective electrodes (more complex), commercial 3-in-1 NPK sensors
GPS Module (Optional)	Captures farm location for region-specific predictions	Crop suitability depends on geographic location	None, or use manual input of farm location
LoRa Module (Optional)	Long-range communication for remote farms	Useful if farm is far from WiFi	WiFi mesh network, NB-IoT modules
Power Supply by Solar Panel	Ensures continuous operation in field	IoT devices need stable power	Rechargeable batteries, UPS for indoor setups

Enclosure / Waterproof Box	Protects electronics in outdoor environment	Prevents damage from weather	Plastic or metal enclosures, weatherproof coating

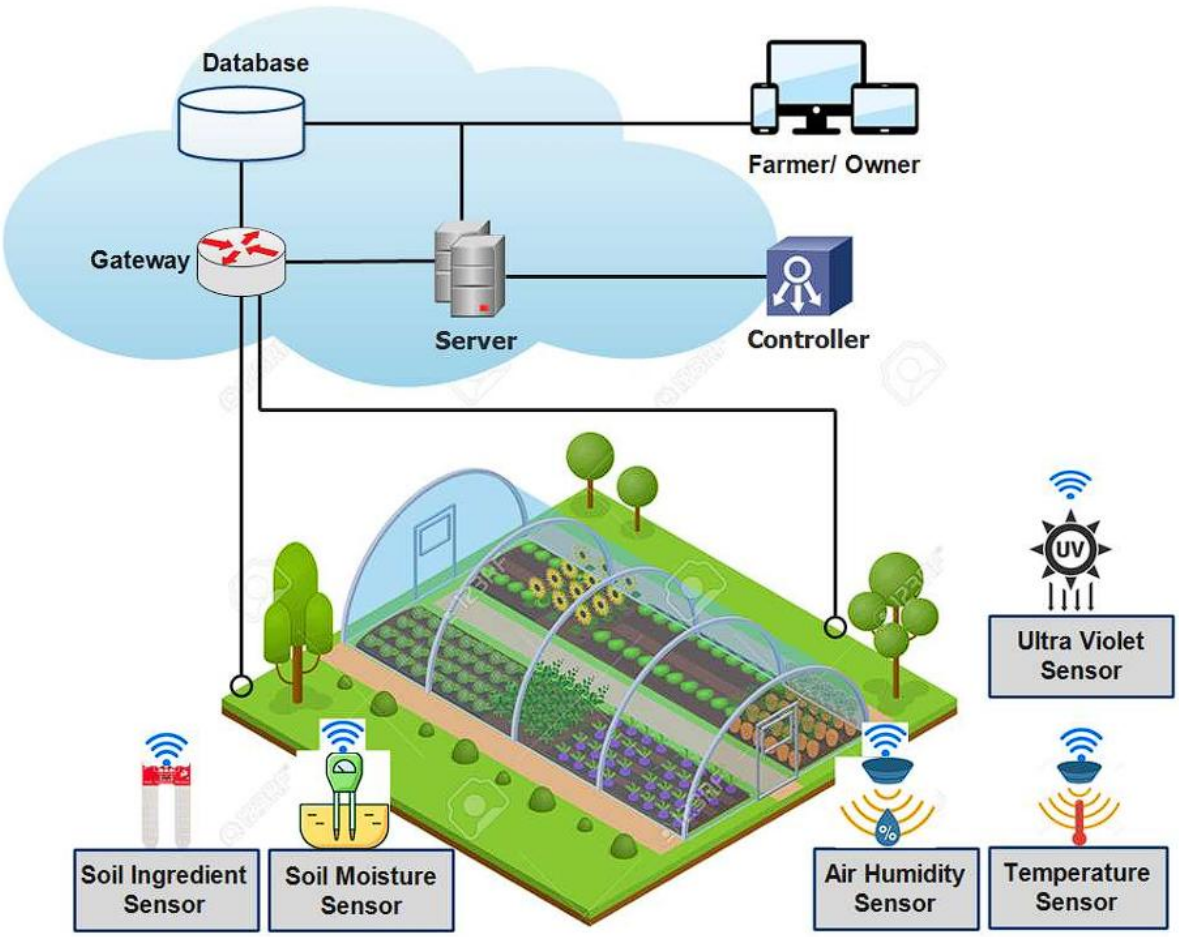

Figure 2: An illustration of IoT application for monitoring farming conditions in a greenhouse [2].

2.3. Feasibility Study

2.3.1 Technical Feasibility

From a technical point of view, this project is completely feasible because it uses technologies and tools that are already familiar to engineering students and are easy to learn within the project timeline. The sensors we are using such as pH, NPK, moisture, temperature, and humidity are all fully compatible with common microcontrollers like ESP32, so the technical setup does not require any advanced industrial hardware. The

programming required involves basic sensor interfacing, simple IoT data transmission, a backend to receive and process the data, and machine learning models that run on Python, which we already use in our academics. Since all the software we need FastAPI, Python, PostgreSQL or Firebase, and dashboard frameworks is free and open source, there is no technical barrier to implementation. The entire architecture (IoT → backend → database → dashboard → predictions) is based on standard practices that we can comfortably implement with available resources. For these reasons, the project is technically feasible and realistic for a final-year major project.

2.3.2 Operational Feasibility

Operationally, the system is simple to run and does not require any specialized training. Once the sensors are installed in the field, the ESP32 automatically sends the readings to the backend, where the machine learning model processes them and the dashboard displays the recommendations. Even someone with basic computer knowledge can operate the dashboard, understand the readings, and follow the recommendations for crop selection, fertilizer usage, and irrigation planning. Maintenance is also easy because individual sensors can be replaced or recalibrated without affecting the whole system, and the software components can be updated or improved at any time. The system can run continuously as long as it has a power source and internet connection, which makes it suitable not only for academic demonstration but also for small-scale real-world deployment. Because of this smooth day-to-day usability, the project is operationally feasible and practical for both students and eventual end-users like farmers

2.3.3 Economic Feasibility

This project is economically feasible as most of the required components are affordable and easily available in Nepal, especially through platforms like Daraz. The School of Engineering also plays an important role here, because the availability of labs, microcontrollers, computer systems, and internet connection significantly reduces our overall cost. The only slightly expensive components in this project are the NPK sensor and the pH sensor, but these are also the most important ones. These two sensors directly determine how accurate and scientific our fertilizer and crop recommendations will be.

Without them, the system cannot give precise output, but with them we can make highly precise and reliable predictions that truly help farmers. This alone justifies the investment in these sensors. Availability of these devices will allow us to build a complete working prototype that can create impact and can also be used for future research works. Overall, the cost of the project is feasible compared to the usefulness and outcomes it delivers.

2.4 Research Methodology

The methodology adopted for building this SSA platform involves an iterative process using intensive reviews to establish a smart farming system. It starts with a clear definition of goals and objectives for the system with the thorough study of papers, articles and existing systems. Then, an IoT based SSA architecture is designed, and finally a prototype of the platform is developed. The system is enhanced and refined over time through occasional updates.

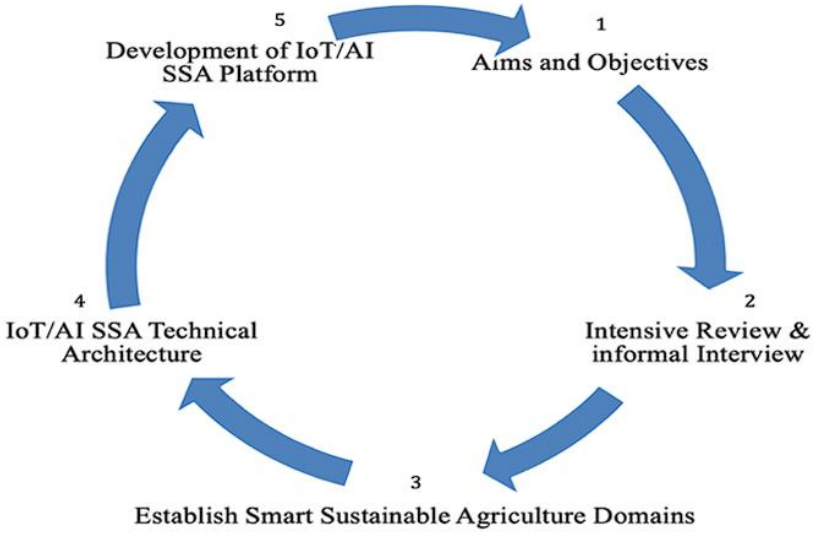

Figure 3: IoT/SSA Research Methodology [1].

3. High Level Design of the System

3.1 System Overview

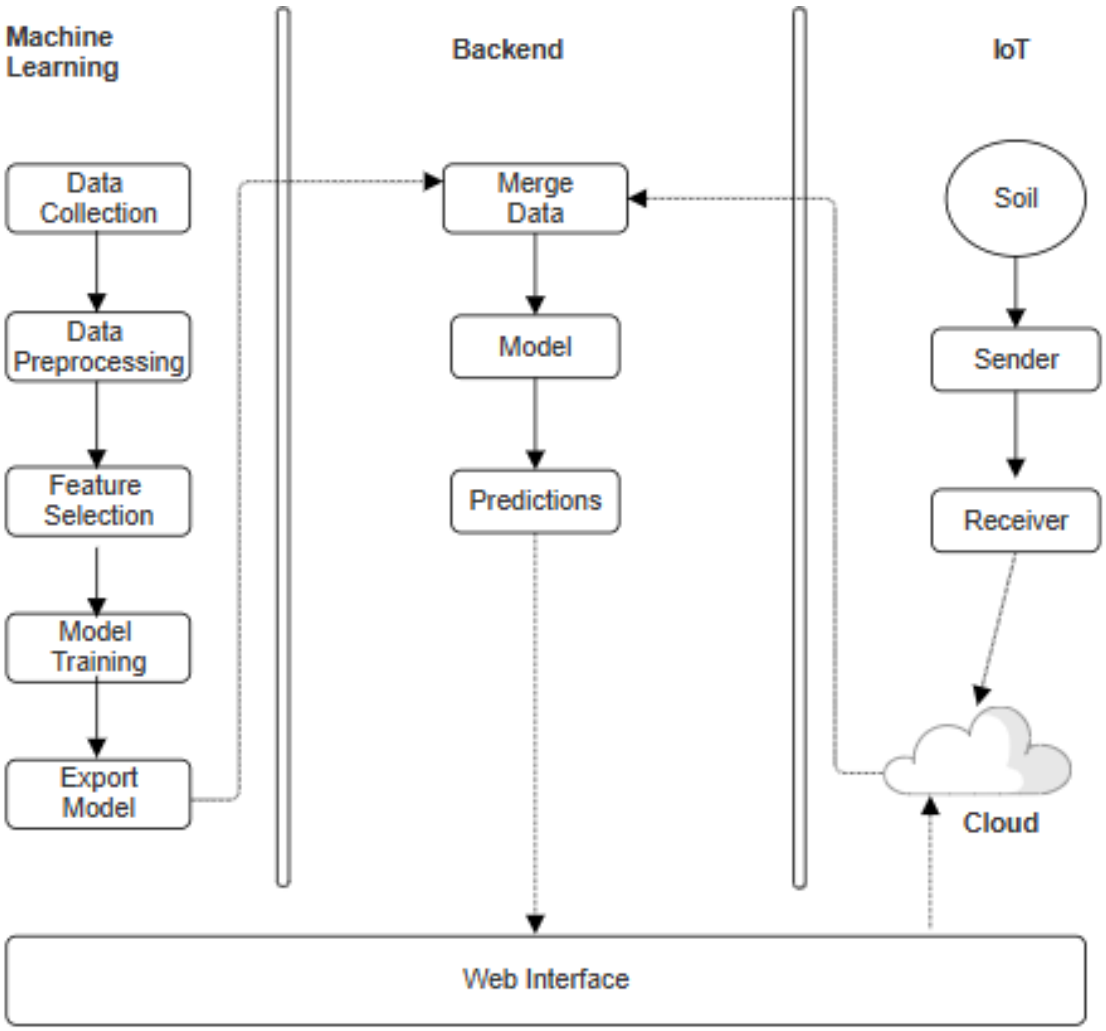

Figure 4: Overall System Block Diagram.

The block diagram above represents the possible architecture of our system. It shows how sensors, controllers, data processing models and user interface interact with each other to make a functional system.

3.2 System Flow Diagram

The flowchart of our system is drawn below:

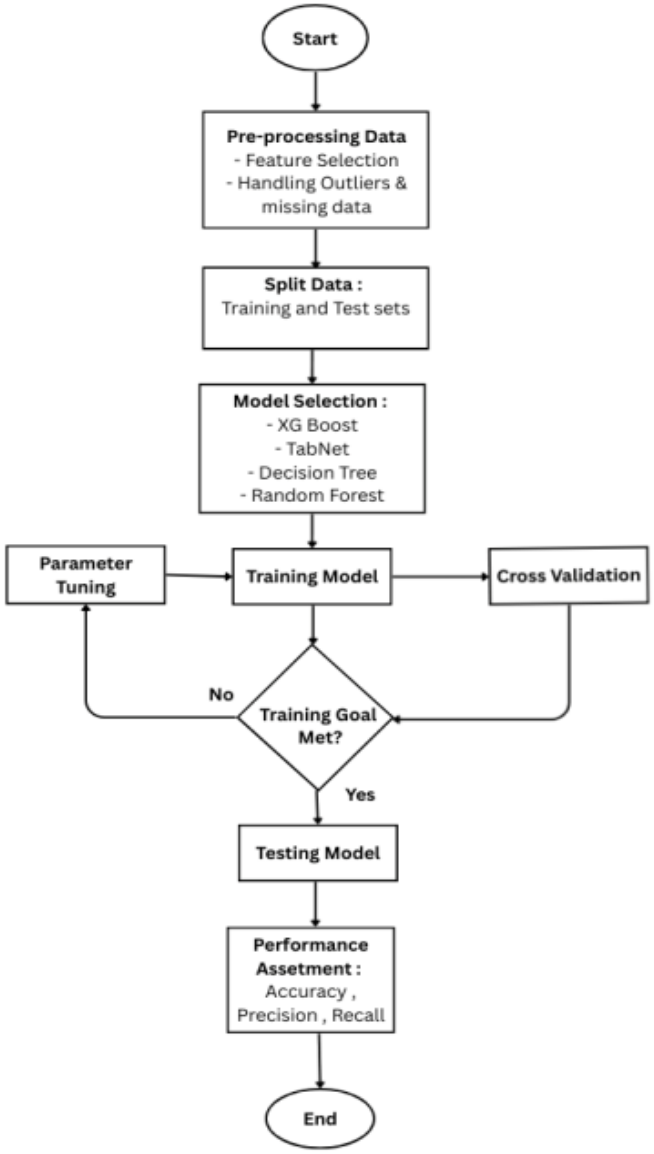

Figure 5: Basic ML workflow of our system.

Sensors take data readings (temperature, humidity, moisture, pH and N, P, K content) from physical surroundings. Then the collected data is sent to backend of the system which then receives and validates the input data and then stores in database. Then the ML model is fed with the data and predictions will be made on crop, fertilizer and irrigation recommendation. The predictions are then stored in database and used to display insights to farmer's dashboard.

4. Algorithm & Mathematical Model

4.1 Random Forest

Random Forest is a machine learning ensemble algorithm that builds many decision trees and combines their outputs to make a final prediction. Random Forest models are highly beneficial for smallholder farms in developing countries due to their ability to handle complex and noisy data, which is common in agricultural contexts. These models excel at capturing nonlinear relationships between variables such as weather patterns, soil characteristics, crop types, and yields, providing accurate predictions and insights for farmers. The ensemble learning approach of Random Forests mitigates overfitting and improves generalization, making them suitable for situations with limited data availability. Following steps are involved in a Random Forest Algorithm:

1. For b=1 to N=B:
    a) Draw a bootstrap sample Z^* of size N from the training data.
    b) Grow a random-forest tree T_b to the bootstrapped data, by recursively repeating the following steps for each terminal node of the tree, until the minimum node size n_{min} is reached.
        i) Select m variables at random from the p variables.
        ii) Pick the best variable/split-point among the m.
        iii) Split the node into two daughter nodes.
2. Output the ensemble of trees $\{T_b\}_1^B$.

To make a prediction at a new point x:

Regression: $\hat{f}_{\mathrm{rf}}^B(x) = \frac{1}{B}\sum_{b=1}^{B} T_b(x)$.

Classification: Let $\hat{C}_b(x)$ be the class prediction of the bth random-forest tree. Then $\hat{C}_{\mathrm{rf}}^B(x) = majority\ vote\ \{\hat{C}_b(x)\}_1^B$.

4.2 TabNet + XAI

TabNet is a deep learning architecture specifically designed for raw and unprocessed tabular data making it ideal for analyzing soil conditions. Unlike traditional models that

often treat all input features equally, TabNet learns to focus selectively on the most relevant ones—such as soil moisture, pH, and nutrient levels—at each decision step. When combined with explainable AI (XAI) methods like SHAP or LIME, TabNet not only provides accurate fertilizer or treatment recommendations but also explains which factors influenced those decisions.

We can use TabNet for classification by following steps:

1. Preprocess the data by cleaning and transforming the features.

2. Split the dataset into training and test sets.

3. Train the TabNet model on the training set.

4. Evaluate the model's performance on the test set.

5. Deploy the model to production so that it can be used for fertilizer recommendation.

5. Expected Outcomes

A fully functional web-based platform will be developed to assist users in accessing legal

- Live dashboard showing temperature, moisture, pH, NPK ratios.
- Crop recommendation with confidence score.
- Daily irrigation schedule (e.g., “Irrigate 25 mm today at 6 AM”).
- Fertilizer suggestion (e.g., “Apply 30 kg Urea/ha”).
- SMS/app alerts or notification: “Moisture low → Starting pump”.
- Accuracy target: ≥92% on crop recommendation, ≥95% irrigation savings vs traditional.

6. Project Timeline

The project will be executed in a phased manner beginning with conceptual research planning and an extensive literature review. This will be followed by hardware setup and

field deployment of sensors for real-time data acquisition. A mathematical and computational model will then be developed, and machine learning algorithms will be trained using simulated datasets before being validated with actual field data. Finally, an intelligent recommendation system will be designed to support farmers in decision-making. Despite the technical and documentation challenges within the limited timeframe, the activities have been systematically scheduled to ensure timely completion as illustrated in the Gantt chart below.

Table 2: Gantt Chart (Feb 2026 - Aug 2026)

Activities	Feb	Mar	Apr	May	Jun	Jul	Aug
Literature Review & Proposal	■	■					
Hardware Procurement & Setup		■	■				
Sensor Integration & Calibration			■	■			
Cloud Setup & Database			■	■			
AI Model Development			■	■	■	■	
Web + Mobile App Development			■	■	■	■	
Field Testing & Validation						■	■
Documentation & Presentation	■				■		■

7. Detailed Budget

The estimated budget required to complete this project, encompassing all anticipated expenses, is outlined below:

Table 3: Budget Breakthrough for Various Titles for Completing Research

Itemize whole budget under following headings		*Total Cost NPR*	**Contribution requested from RIDC**
A. Equipment/ materials			
1.	Soil Sensors (DHT22, NPK, PH, etc.)	16,000	16,000
2.	Solar power supply	4,500	4,500
3.	Microcontroller (ESP32)	1,500	1,500
4.	Breadboard and jumping wires	500	500
B. Implementation and Deployment cost			
1.	Use of Cloud services for storage and compute	2,000	2,000
2.	Discussion with Industry Expert Personnel	2,000	2,000
C. Other cost			
1.	Literature review (Email, Internet)	*	*
2.	Group discussion with Domain expert	*	*
3.	Draft and Final Report Printing, Copying and Binding cost	1,000	1,000
4.	Communication (Email and Internet, Telephone)	500	500
5.	Miscellaneous and Transportation cost	2,000	2,000
Total (NPR): Thirty Thousand Only		30,000/-	30,000/-

8. References

[1] E. Alreshidi, "Smart Sustainable Agriculture (SSA) Solution Underpinned by Internet of Things (IoT) and Artificial Intelligence (AI)," Int. J. Adv. Comput. Sci. Appl., vol. 10, no. 5, pp. 93–102, 2019.

[2] V. K. Quy, N. V. Hau, D. V. Anh, N. M. Quy, N. T. Ban, S. Lanza, G. Randazzo, and A. Muzirafuti, "IoT-Enabled Smart Agriculture: Architecture, Applications, and Challenges," Appl. Sci., vol. 12, no. 7, p. 3396, 2022, doi: 10.3390/app12073396.

[3] "Optimizing Agricultural Efficiency: A Fusion of IoT, AI, Cloud Computing, and Wireless Sensor Network," Preprint, SSRN Electron. J., 2024, doi: 10.2139/ssrn.4789232.

[4] A. S. Ibrahim et al., "AI-IoT based smart agriculture pivot for plant diseases detection and treatment," Sci. Rep., vol. 15, no. 1, p. 16576, Dec. 2025, doi: 10.1038/s41598-025-98454-6.

[5] Hossain, I., et al. (2022). IoT Based Smart Irrigation System on Renewable Energy. International Journal of Innovative Science and Research Technology, 7(6), 53–59.

[6] Kim, J., et al. (2023). Real-Time Smart Farming Using IoT and AI Models. Agriculture, 15(5), 554. https://doi.org/10.3390/agriculture15050554

[7] Reddy, K. P., & Ghosh, A. (2021). Performance Analysis of IoT Based Smart Agriculture Systems. Wireless Personal Communications, 120, 2569–2591. https://doi.org/10.1007/s11277-021-09462-4

[8] Zhang, X., et al. (2025). Edge and Cloud Enabled Smart Farming Framework. IEEE Access. https://doi.org/10.1109/ACCESS.2025.3594162

[9] Liu, Y., & Singh, P. (2024). Drivers and Barriers to IoT Adoption in Agriculture. Heliyon, e25112. https://doi.org/10.1016/j.heliyon.2024.e25112

[10] Ahmed, S., & Khan, R. (2021). Machine Learning Approaches for Precision Agriculture. IEEE Access. https://doi.org/10.1109/ACCESS.2021.3138160